

PRATEEK DWIVEDI

प्रतीक द्विवेदी

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RESEARCH INTERESTS

My research focuses on the theoretical foundations of computer science, with particular emphasis on **Algebraic Complexity Theory** and **Computational Complexity**.

EMPLOYMENT

IT University of Copenhagen
Postdoctoral Researcher
Theoretical Computer Science Section
Adviser: [Prof Nutan Limaye](#)

Copenhagen, Denmark
2024 -

EDUCATION

Indian Institute of Technology Kanpur
Doctor of Philosophy
Computer Science and Engineering
Thesis: *Treading the Borders for Explicitness, Circuit Factoring, and Identity Testing* [[pdf](#)]
Adviser: [Prof Nitin Saxena](#)

Kanpur, India
2018 - 2025

KCC Institute of Technology and Management
Bachelor of Technology
Computer Science and Engineering

Gautam Buddha Nagar, India
2013 - 2017

PUBLICATIONS

Lower Bounds in Algebraic Complexity via Symmetry and Homomorphism Polynomials
with Benedikt Pago and Tim Seppelt
Submitted, 2025

2025

On Closure Properties of Read-Once Oblivious Algebraic Branching Programs
with Robert Andrews, Jules Armand, Magnus Hansen, Nutan Limaye, Srikanth Srinivasan, and Sébastien Tavenas
17th Innovations in Theoretical Computer Science (ITCS), 2026

2025

A primer on the closure of algebraic complexity classes under factoring
with C. S. Bhargav and Nitin Saxena
To appear in the proceedings of Recent Trends in Computer Algebra (RTCA), 2025

2025

Monotone bounded-depth complexity of homomorphism polynomials
with C. S. Bhargav, Shiteng Chen, Radu Curticapean
50th International Symposium on Mathematical Foundations of Computer Science (MFCS), 2025

2024

Learning the coefficients: A presentable version of border complexity and applications to circuit factoring
with C. S. Bhargav and Nitin Saxena
56th Annual ACM Symposium on Theory of Computing (STOC), 2024

2024

Lower bounds for the sum of small-size algebraic branching programs with C. S. Bhargav and Nitin Saxena <i>19th Annual Conference of Theory and Applications of Models of Computation (TAMC), 2024</i> <i>Invited in the special issue of Theoretical Computer Science (Theor. Comput. Sci.)</i>	2024
Demystifying the border of depth-3 algebraic circuits with Pranjal Dutta and Nitin Saxena <i>62nd Annual Symposium on Foundations of Computer Science (FOCS)</i> <i>Invited in the special issue of SIAM Journal on Computing (SICOMP)</i>	2021
Deterministic identity testing paradigms for bounded top-fanin depth-4 circuits with Pranjal Dutta and Nitin Saxena <i>36th Computational Complexity Conference (CCC)</i> <i>The full version is currently under review in Theory of Computing (ToC)</i>	2021

ACADEMIC TALKS AND SEMINARS

Non-Closure properties in algebraic complexity <i>IIT Delhi, India</i>	2025
Monotone Bounded-Depth Complexity of Homomorphism Polynomials <i>ARCO, Malmö University, Sweden</i>	2024
Presentable Version of Border Complexity and Applications to Circuit Factoring • <i>Ulm University, Germany</i> • <i>Welcome talk at the Theory Group, ITU, and BARC, KU</i>	2024 2024
Treading the Border Complexity and Identity Testing Paradigms <i>Thesis Defense, IIT Kanpur</i>	2022
Deterministic identity testing paradigms for bounded top-fanin depth 4 circuits <i>Invited Talk, CS Theory Seminar at Georgetown University</i>	2021
Deterministic identity testing paradigms for bounded top-fanin depth-4 circuits • <i>Workshop on Algebra and Computation (WAC), Göteborg</i> • <i>7th Workshop on Algebraic Complexity Theory (WACT), Warwick</i> • <i>CS Theory Seminar at Georgetown University</i> • <i>Conference Presentation, (CCC)</i>	2023 2023 2021 2021
Information-Theoretic And Algorithmic Thresholds For Group Testing <i>PhD Comprehensive Evaluation, IIT Kanpur</i>	2020
On Approximative Closure of Algebraic Complexity Classes <i>SIGTACS, IIT Kanpur</i>	2019

SHORT TERM VISITS

50th International Symposium on Mathematical Foundations of Computer Science (MFCS) <i>Warsaw, Poland</i>	2025
57th Annual ACM Symposium on Theory of Computing (STOC) <i>Prague, Czech Republic</i>	2025

Université Savoie Mont Blanc, France <i>Host: Prof. Sébastien Tavenas</i>	2025
Tata Institute of Fundamental Research (TIFR), Mumbai <i>Host: Prof Mrinal Kumar</i>	2025
Ulm University, Germany <i>Host: Prof Thomas Thierauf</i>	2024
Algorithmic Research-Cooperation around Oresund (ARCO) <i>Malmö University, Sweden</i>	2024
Summer of Counting and Algebraic Complexity <i>IT University of Copenhagen (ITU) in Copenhagen, Denmark</i>	2023
Workshop on Algebra and Computation (WAC) <i>Chalmers University of Technology, Göteborg</i>	2023
7th Workshop on Algebraic Complexity Theory (WACT) <i>The University of Warwick, UK</i>	2023
2nd Swiss Winter School on Theoretical Computer Science <i>Jointly organized by EPFL and ETH Zurich in Zinal</i>	2023
Chennai Mathematical Institute (CMI) <i>Host: Prof Partha Mukhopadhyay</i>	2022
62nd Annual Symposium on Foundations of Computer Science (FOCS) <i>Denver, Colorado - Attended Virtually</i>	2022
36th Computational Complexity Conference (CCC), 2021. <i>Toronto, Ontario, Canada - Attended Virtually</i>	2021
48th International Colloquium on Automata, Languages, and Programming (ICALP) <i>Glasgow, Scotland - Attended Virtually</i>	2021
52nd ACM Symposium on Theory of Computing (STOC) <i>Chicago, US - Attended Virtually</i>	2020
Workshop on Algebraic Complexity Theory <i>ICTS, Bengaluru</i>	2019

ACADEMIC ACHIEVEMENTS

Invited to participate in Summer of Counting and Algebraic Complexity at ITU Copenhagen <i>Full financial support</i>	2023
Selected to attend Swiss Winter School on Theoretical Computer Science <i>Partial financial support</i>	2023
Recipient of financial support from Microsoft Research India and IARCS-ACM India <i>To attend an academic conference</i>	2022
Ranked 23 in Joint Entrance Screening Test (JEST)	2018
Ranked 1179 in Graduate Aptitude Test in Engineering (GATE) Computer Science <i>Among approximately 1,00,000 candidates</i>	2017
Presented undergraduate project at university-level Science and Technology Conference.	2017
Academic Excellence Award <i>KCC ITM, India</i>	2014

TEACHING AND MENTORING EXPERIENCE

Undergraduate Project, ITU Copenhagen

2025

Mentoring Thøger Bro and Emil Andreas Sondum Jointly with Prof Riko Jacob

Head Tutor at CSE, IIT Kanpur

Responsible for practical assignments, Lab Exam, and all the course-related logistical support

- Data Structures and Algorithms

2024

Mathematics for Computer Science, IIT Kanpur

Adhyayan 2023, Summer School. Jointly with C. S. Bhargav

Tutor at CSE, IIT Kanpur

- Introduction to Computing
- Data Structures and Algorithms

2022

2019

Teaching Assistant at CSE, IIT Kanpur

- Computational Complexity
- Modern Cryptology
- Mathematics for Computer Science
- Data Structures and Algorithms
- Introduction to Computing

2021

2021

2020

2018

2018

Teaching Assistant in Massive Online Courses

- Introduction to Cryptography | eMasters, IIT Kanpur
- Randomized methods in Computational Complexity | NPTEL
- Arithmetic Circuit Complexity | NPTEL
- Modern Algebra | NPTEL

2022

2021

2020

2019

PROFESSIONAL CONTRIBUTIONS

Reviewer for Conferences

STACS 2026, STOC 2025, FOCS 2025, ITCS 2025, STOC 2025, STOC 2024, FSTTCS 2024, FOCS 2023, ISSAC 2023

STOC 2023, CCC 2022, ITCS 2022, SODA 2021, FOCS 2021

Reviewer for journals

SIAM Journal on Computing

Seminar Organiser at CSE, IIT Kanpur

Regularly organise talks and seminars in a department special interest group (SIGTACS)

PERSONAL DETAILS

Name

Prateek Dwivedi (प्रतीक द्विवेदी)

Gender and Pronouns

Male and He/Him.

Permanent Address

A106, Bulland Heights, Crossings Republik
Ghaziabad (U.P) - 201016, India

Date and place of birth	21 st Feb 1996, Lucknow, India
Nationality	Indian (भारतीय)
Contact	+91 8800 291 032 +45 31 81 36 18

REFERENCES

Nitin Saxena	Nutan Limaye	Nikhil Balaji
Professor	Professor	Assistant Professor
IIT Kanpur	ITU Copenhagen	IIT Delhi
nitin@cse.iitk.ac.in	nuli@itu.dk	nbalaji@cse.iitd.ac.in